

Coexisting equilibrium states without a phase transition

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Conjecture 00000008476

Result. Nonuniqueness of equilibrium states does not imply nonanalyticity of the pressure defined in the source.

A compact dynamical system. Take the discrete compact space $X = \{0, 1\}$, the continuous map $T = \text{id}_X$, and the continuous potential $f = 0$. Each point mass δ_a is an invariant ergodic probability measure: every invariant measurable set has δ_a -measure zero or one. The two measures δ_0 and δ_1 are distinct.

The actual measure entropy. For any probability measure μ on X and any finite measurable partition $\mathcal{P}$, its Shannon entropy is

$$H_\mu(\mathcal{P}) = - \sum_{A \in \mathcal{P}} \mu(A) \log \mu(A), \quad 0 \log 0 := 0.$$

Since T is the identity, the common refinement $\mathcal{P}^{(n)} = \bigvee_{j=0}^{n-1} T^{-j} \mathcal{P}$ is exactly $\mathcal{P}$ for every $n \geq 1$. Therefore

$$h_\mu(T, \mathcal{P}) = \lim_{n \rightarrow \infty} \frac{H_\mu(\mathcal{P}^{(n)})}{n} = 0, \quad h_\mu(T) = \sup_{\mathcal{P}} h_\mu(T, \mathcal{P}) = 0.$$

The supremum is over all finite measurable partitions. Thus this is the Kolmogorov–Sinai entropy, not an assigned zero value.

Pressure and maximizers. For every real t and every ergodic invariant probability measure μ ,

$$\int_X f d\mu + t h_\mu(T) = 0.$$

The family of such measures is nonempty, so the genuine supremum in the source is

$$P(t) = \sup_{\mu \text{ ergodic, invariant}} \left(\int_X f d\mu + t h_\mu(T) \right) = 0.$$

This function is real analytic at every real parameter. Nevertheless, both distinct measures δ_0 and δ_1 attain this supremum for every t . Consequently equilibrium states are nonunique at every parameter, while there is no phase transition at any parameter under the source’s explicit definition as a nonanalytic point.

Scope and formalization. Neither source version requires transitivity, mixing, or a hypothesis ensuring that distinct maximizing states yield different pressure branches. Their unrestricted equivalence therefore includes this example. We do not dispute existence or uniqueness theorems with additional dynamical hypotheses.

Lean uses actual measures, ergodicity, Bochner integrals, and real analyticity. All finite partitions of the two-point space are encoded by their equivalence relations; atoms are their quotient classes. The iterated refinement is the infimum of the pulled-back equivalence

relations. The proof computes every partition entropy rate by a genuine limit, then the supremum over all partitions, the full pressure supremum, and two distinct actual maximizing probability measures. A limit superior is used to define partition rates, and the proved ordinary limit is zero, so it agrees with the usual entropy rate.